Project Description

This project will result in the build of a centralized log monitoring server using Splunk Enterprise on the Ubuntu Linux virtual machine (Shea_134) created in the proxmox environment. The goal is to demonstrate how the Splunk platform can collect, search, and visualize system and application logs for monitoring and troubleshooting.

By configuring Splunk to collect logs from the Ubuntu VM (system and authentication logs) and web server logs generated by an Apache HTTP Server, we will see how implementing logging for a system can improve visibility by having visual dashboards and alerts that help identify system events/errors, user activity, and security events on the network.

The dashboards and alerts were completed to demonstrate how this platform can be beneficial in a real world environment.

Objectives

- Run an Apache2 HTTP server on the linux vm
- Download and install Splunk Enterprise onto the vm
- Configure Splunk to run the information from the various log files
- Configure dashboards in Splunk to view the traffic as one would in the real world
- Configure alerts for the user in the Splunk dashboard

Deployment Approach

I configured my machine to have the two CPU cores with 8 GB of memory and 64 GB of storage as recommended. I deployed Splunk Enterprise on the Ubuntu machine in Proxmox. The virtual machine will need to have the desktop LightDM installed in order for the user to access the Apache2 webpage to simulate user activity for the logs. Apache2 will be accessible through port 8000.

I downloaded the Linux Debian version from the Splunk website after creating a profile. After the packages are showing they have successfully installed through the terminal, I verified they are working by going to the Splunk login page that is hosted on the vm at http://shea134. The Splunk server will create logs that will be stored in the “main” index and analyzed through searches, dashboards, and alerts. The alerts will be configured by the administrators. The log files being collected to populate into the dashboard are as follows:

Log File	Source Type	App Context	Index
`/var/log/syslog`	`syslog`	Search & reporting	Main
`/var/log/auth.log`	`linux_secure` (or `syslog`)	Search & reporting	Main
`/var/log/apache2/access.log`	`access_combined`	Search & reporting	Main
`/var/log/apache2/error.log`	`apache:error`	Search & reporting	Main

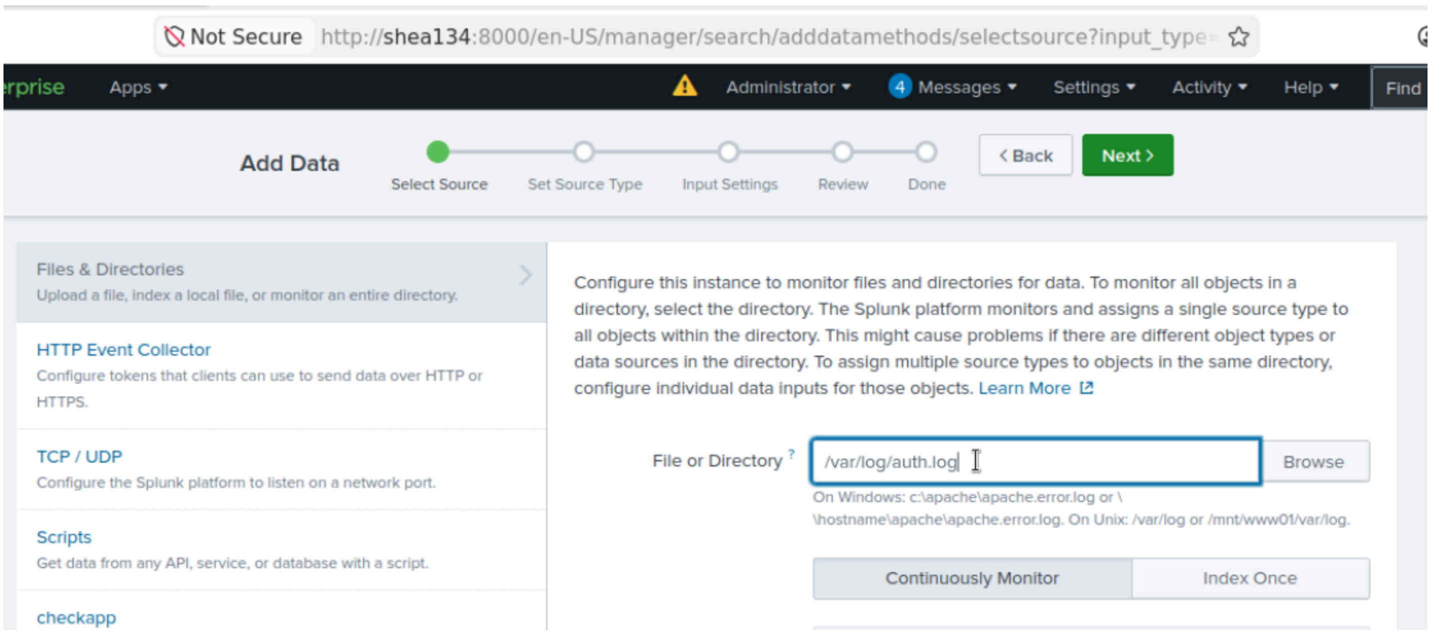
Alert Automation

For this project, I have created alerts that will trigger in Splunk if the Apache webpage generates more than 3 HTTP errors in the 5 minutes. In a real world scenario, I would advise the department to create a shared email. This would notify admins instead of waiting on the admin to login and see the alerts.

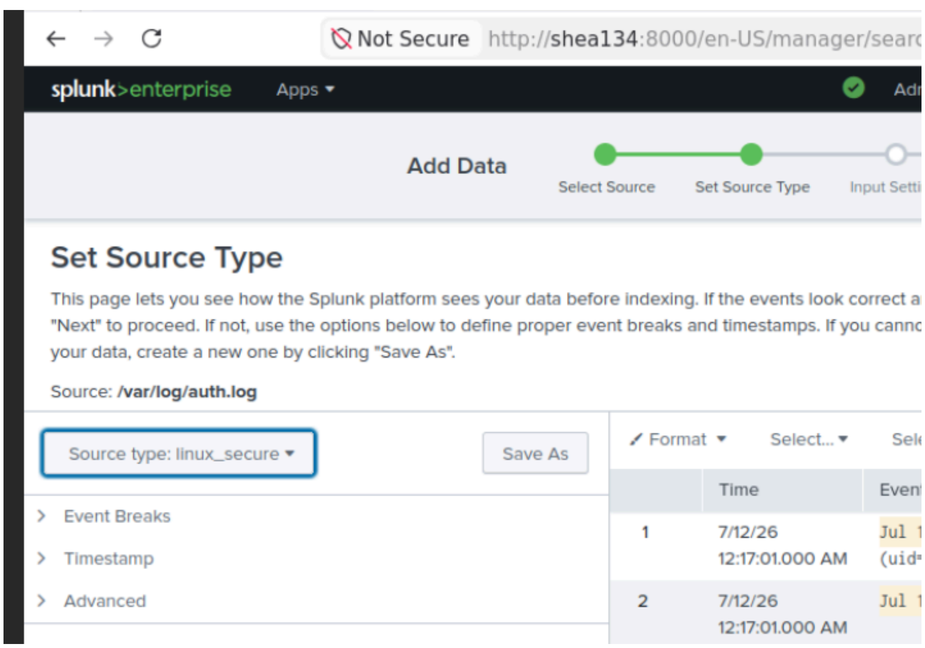
In addition to the automated alerts of the HTTP 404 status codes, the dashboard performs automated monitoring of log files. This was created through the use of data inputs.

Results - Automation

The **Data inputs** were configured in order for the logs to populate in our dashboards. This was done through Settings > Data > Data Inputs > Select **Add New** for the **Files & Directory** type

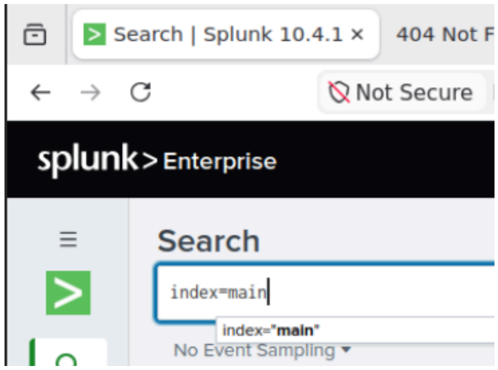


Using the **/var/log/auth.log** file as an example

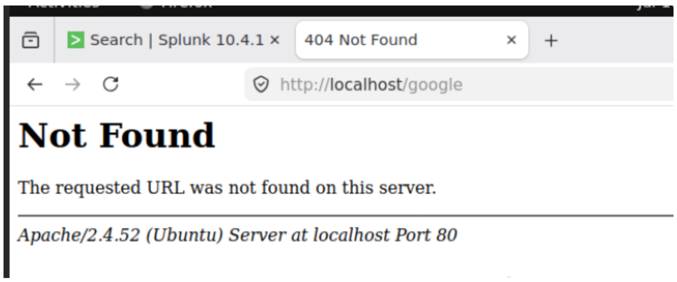


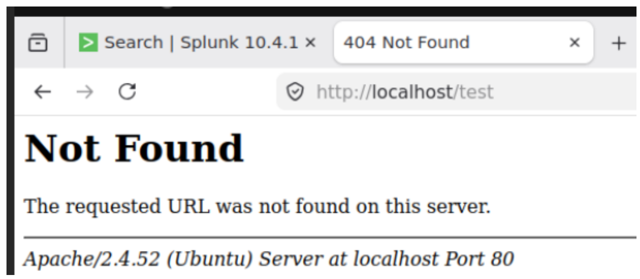
For this file, I changed the source type to **linux_secure** before I could continue

Once the files are entered into the system, we will go to the **Apps** section of Splunk, then the **Search & Reporting** tab to the left.

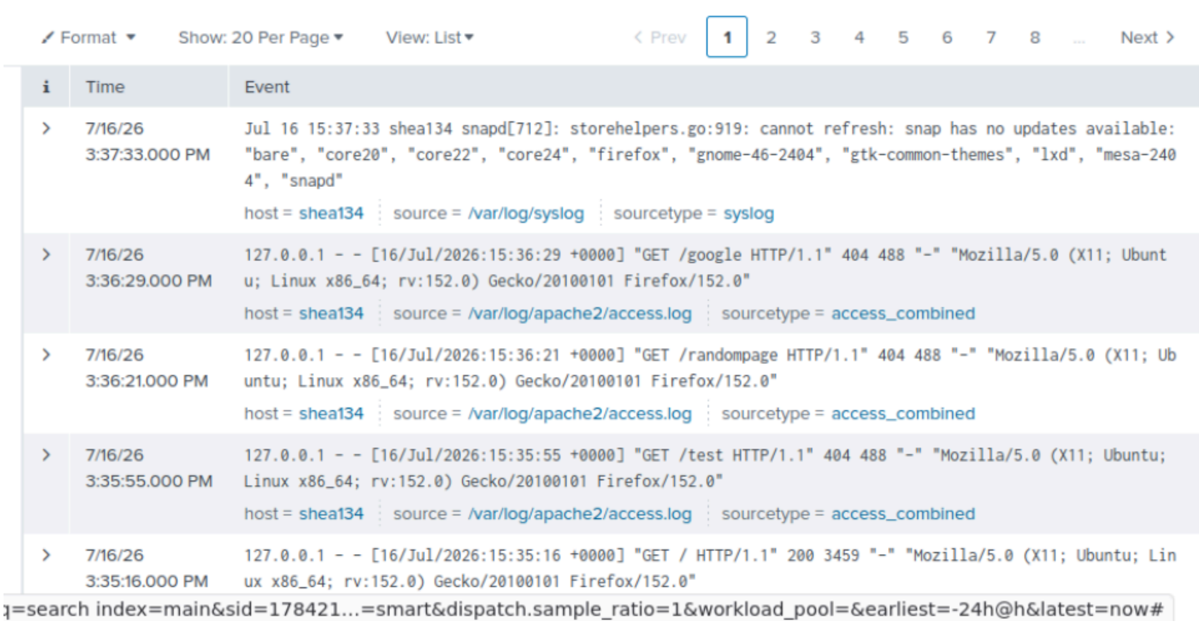


I will need to run **index=main** in the search bar

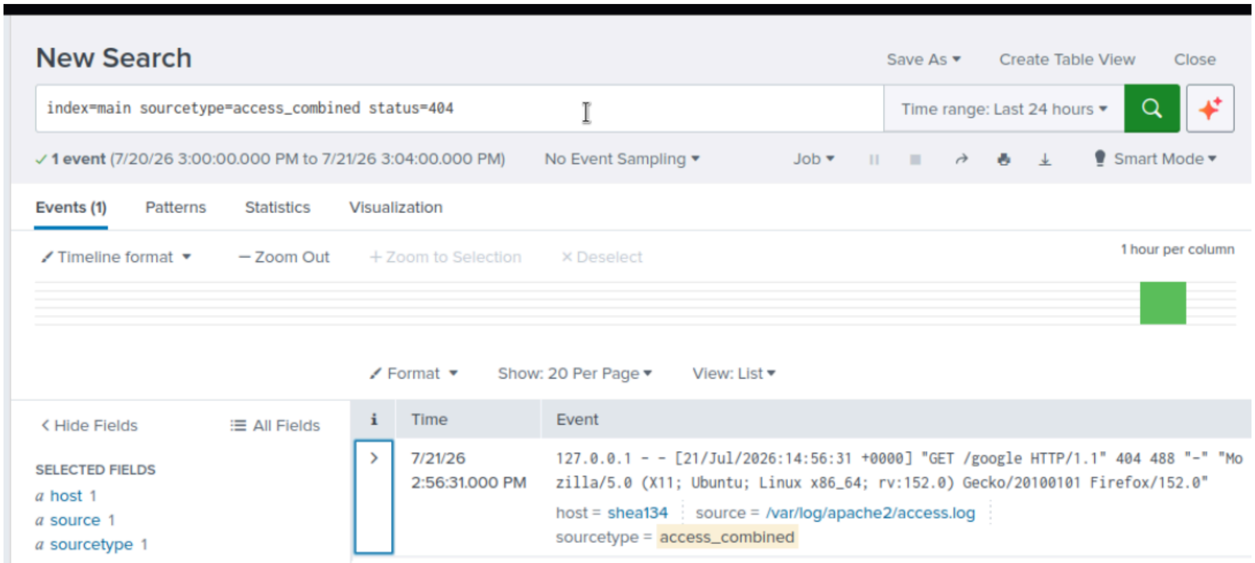




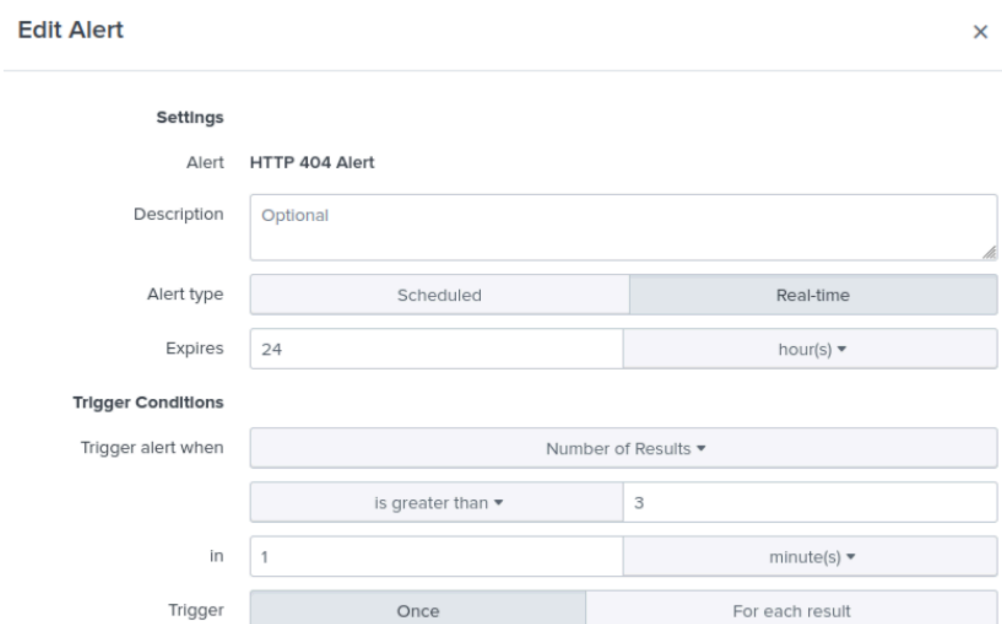
To generate some logs, I will search using the http://localhost/ to search using the apache2 page.



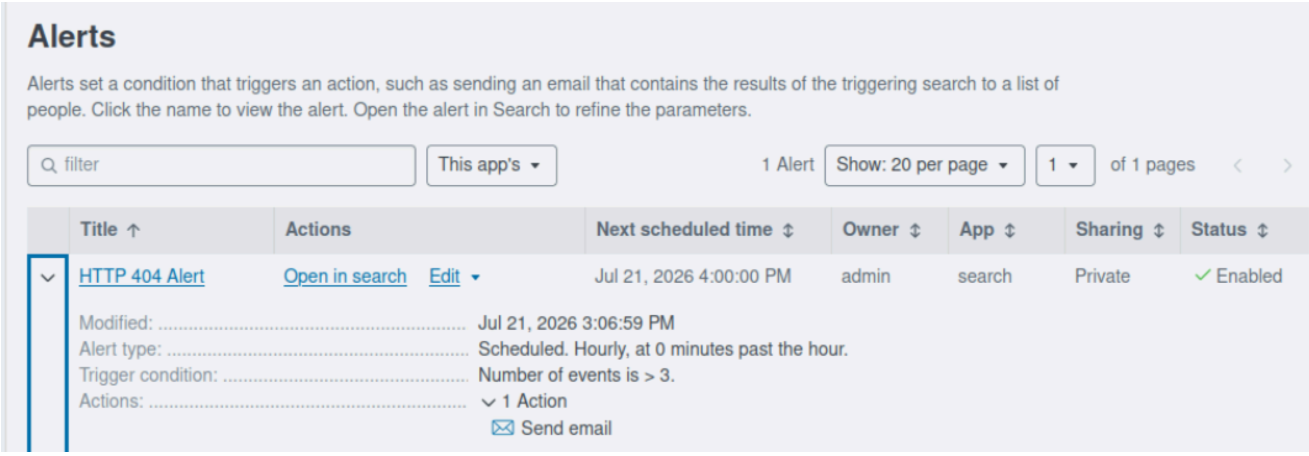
After hitting enter, we start to see some logs show up. As we drill into an event, we can see that I tried to search for **google** and got a 404 error



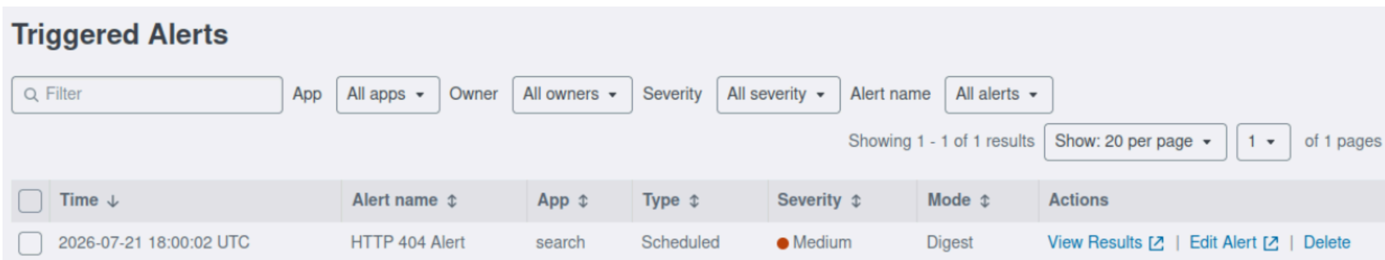
Create a search for **index=main sourcetype=access_combined status=404**. This will pull the 404 errors from the Apache server that are created in the main. This was saved as an alert.



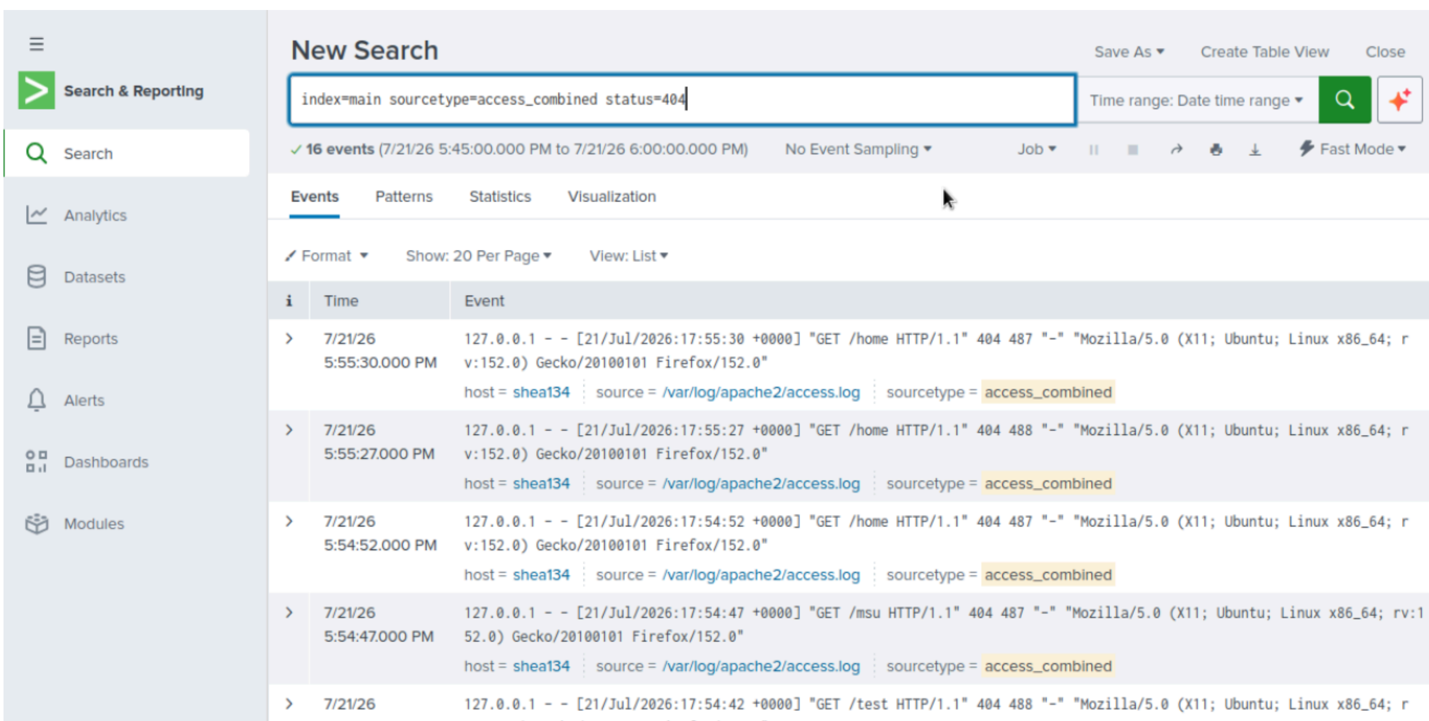
Naming the alert and telling it to run in real time



Configured alert for the 404 HTTP status code



Triggered alert that was automated to the dashboard after simulating some 404 HTTP status codes



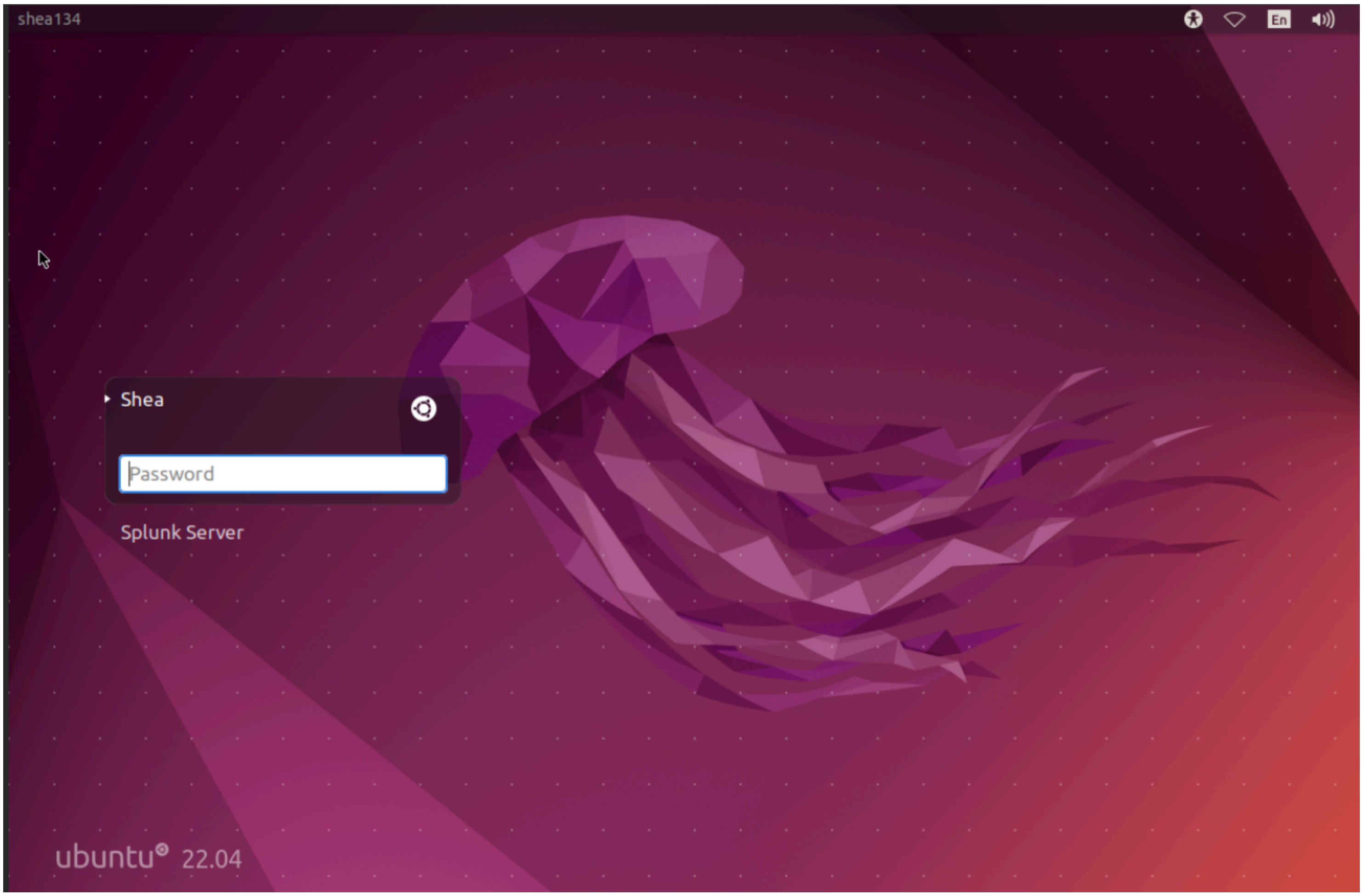
Results of the alert that was triggered through the automation in the dashboard

Cloud Platform

Deployed on the Ubuntu Linux virtual machine 134 created in the course's Proxmox environment. Because this platform was installed as the console only, the LightDM dashboard was installed through the terminal. The desktop version was needed in order to simulate user interactions with the Apache2 HTTP webpage hosted on the server.

Commands to install the desktop:

- sudo apt update // refreshing packages and versions
- sudo apt install xfce4 lightdm firefox -y // installs the lightdm desktop and firefox
- sudo reboot // reboot to switch to the desktop

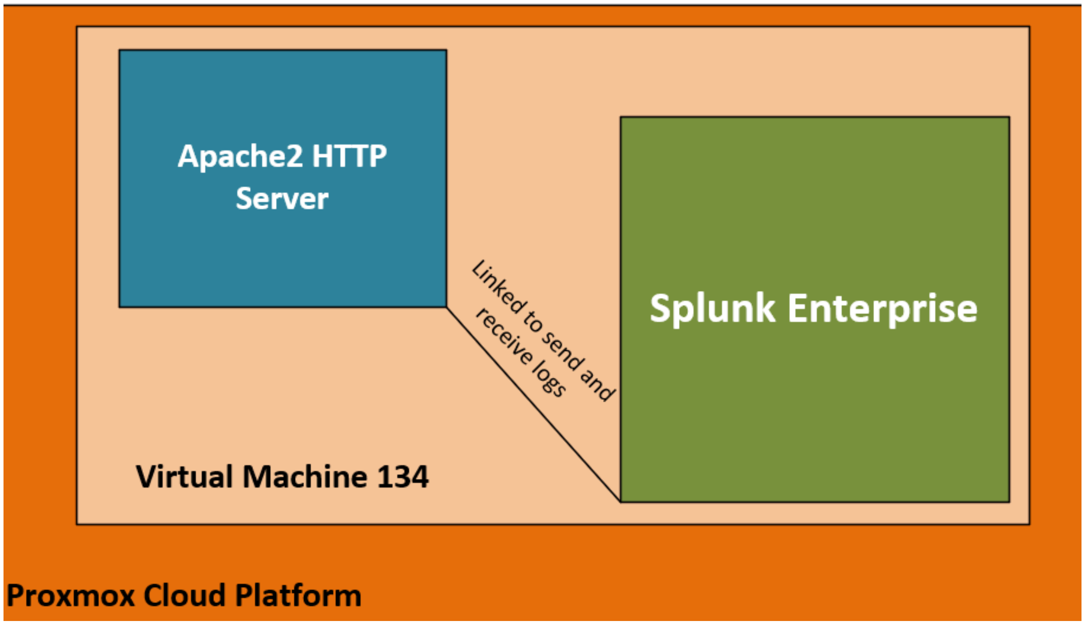


LightDM Desktop

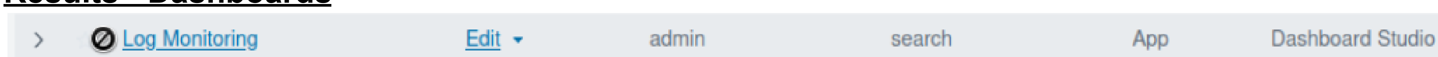
Architecture Diagram

The Splunk Enterprise and Apache2 HTTP server have been linked through the configuration of logs. Both are hosted on the 134 virtual machine that lives in the Proxmox Cloud Platform.

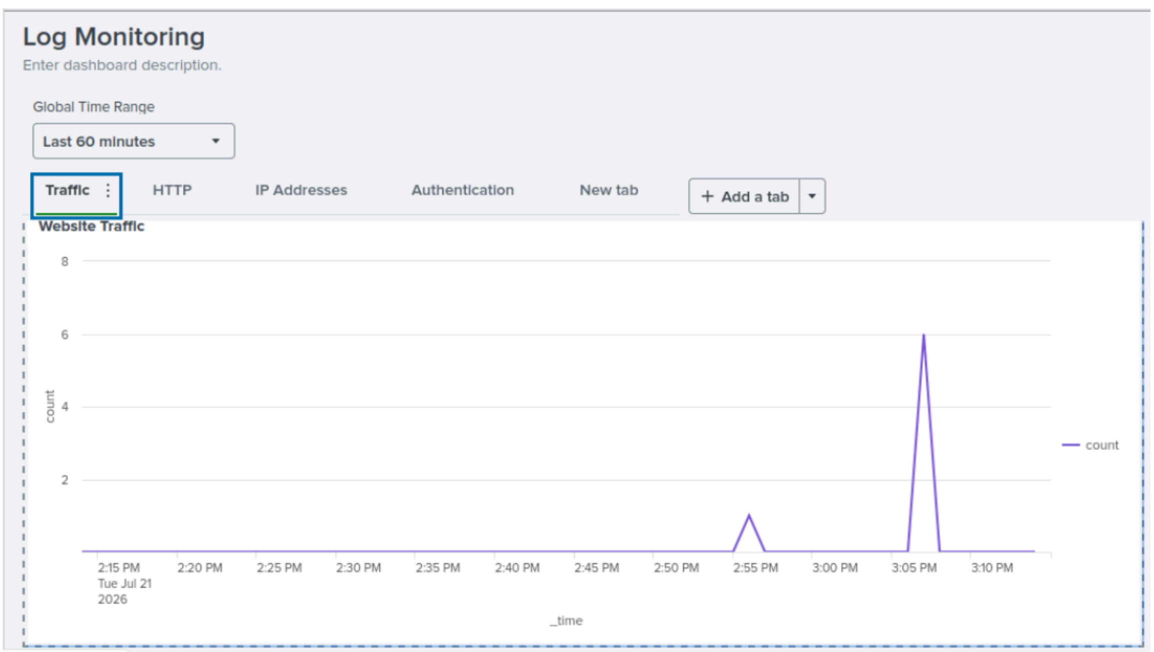
The Apache server writes to the splunk log "/var/log/apache2/*.log" and then is monitored with the splunk enterprise data inputs. Splunk then indexes the data we configured in 'Data Inputs' and will display the data in graphs in the form of a dashboard.



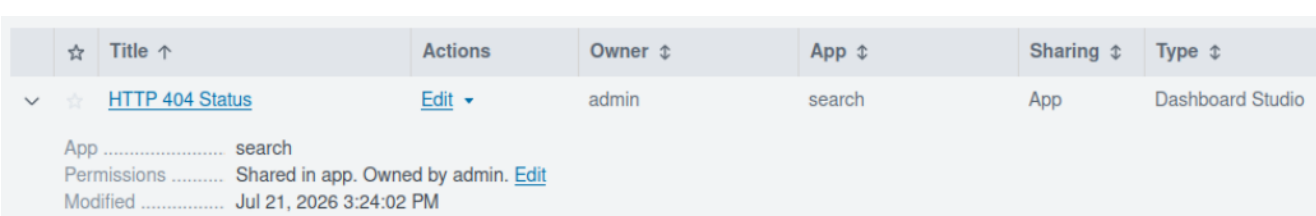
Results - Dashboards



Existing dashboard link under the **Dashboards** tab



Log monitoring dashboard with panels



Configured Dashboard for HTTP 404 status

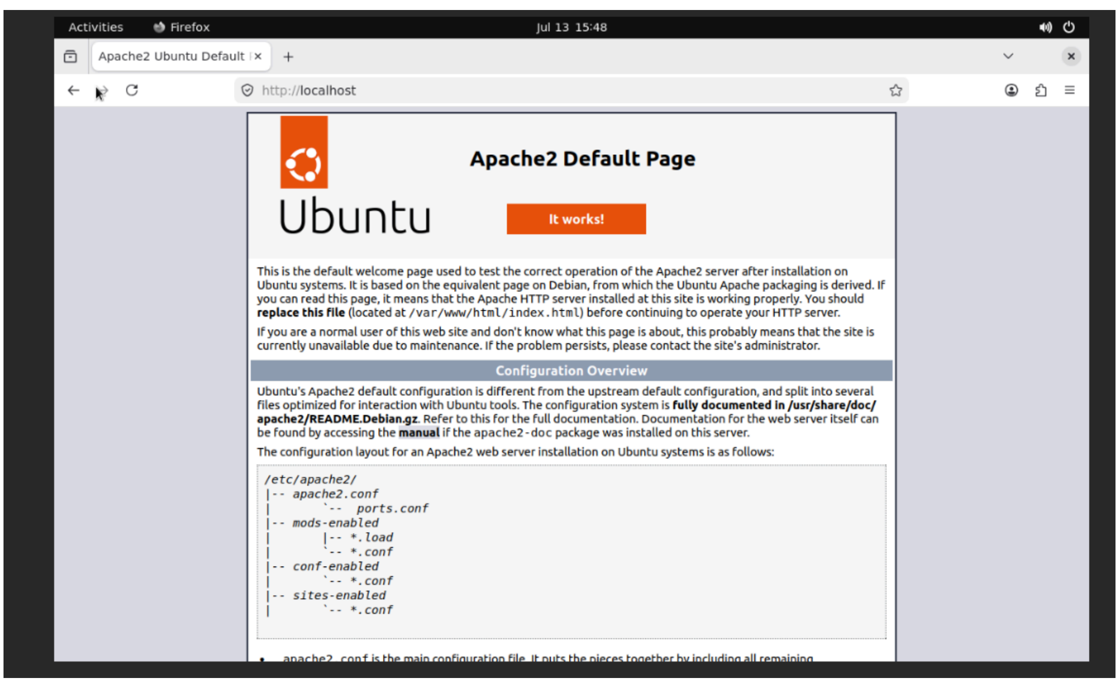
HTTP 404 Status Dashboard

<u>Starting Apache2 HTTP Server</u>

These commands will install Apache2 and tell it to start up upon starting the virtual machine without needing user intervention.

<u>Commands to start the Apache2 service:</u>

> sudo apt install apache2 -y
> sudo systemctl enable apache2
> sudo systemctl start apache2



Working http page http://localhost

Installing Splunk

For the Linux machine, we will need to download the Linux version of Splunk. Using the wget link, I was able to download and install the software through the terminal.

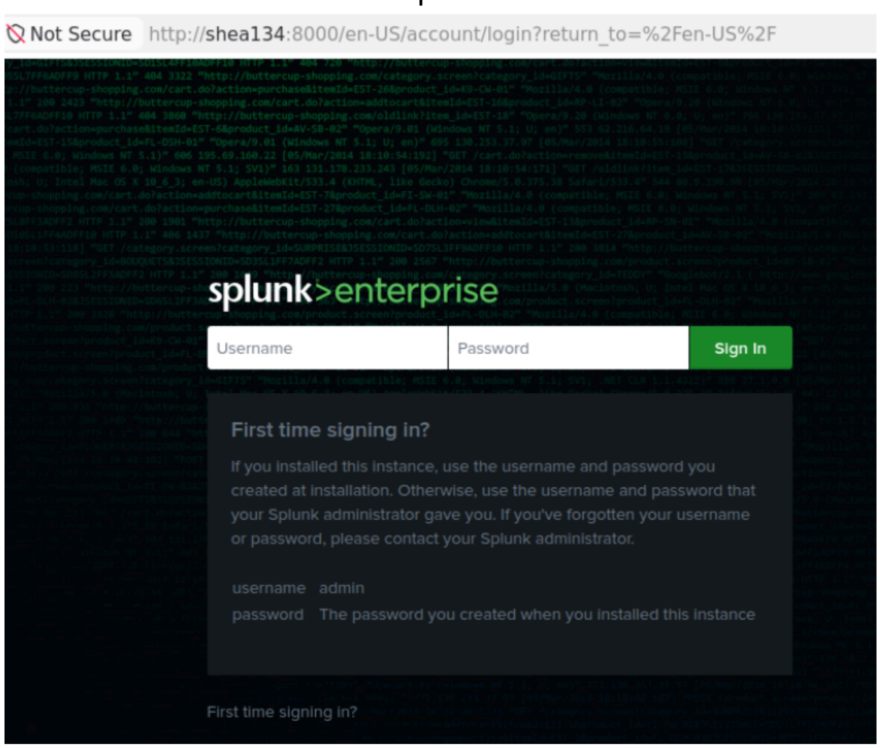
```
ubuntu134@shea134:~/Downloads$ wget -O splunk-10.4.1-5a009d941268-linux-amd64.deb "https://download.splunk.com/products/splunk/releases/10.4.1/linux/splunk-10.4.1-5a009d941268-linux-amd64.deb"
--2026-07-16 14:24:23--  https://download.splunk.com/products/splunk/releases/10.4.1/linux/splunk-10.4.1-5a009d941268-linux-amd64.deb
Resolving download.splunk.com (download.splunk.com)... 54.192.100.74, 54.192.100.32, 54.192.100.68, ...
Connecting to download.splunk.com (download.splunk.com)|54.192.100.74|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1322956776 (1.2G) [binary/octet-stream]
Saving to: ‘splunk-10.4.1-5a009d941268-linux-amd64.deb’

-linux-amd64.deb              20%[============>                    ] 261.84M  11.2MB/s    eta 92s
```

Downloaded the splunk package through the terminal

```
ubuntu134@shea134:~/Downloads$ sudo dpkg -i splunk*.deb
[sudo] password for ubuntu134:
Selecting previously unselected package splunk.
(Reading database ... 172776 files and directories currently installed.)
Preparing to unpack splunk-10.4.1-5a009d941268-linux-amd64.deb ...
verify that this sytem has all the commands we will require to perform the preflight step
no need to run the splunk-preinstall upgrade check
Unpacking splunk (10.4.1) ...
```

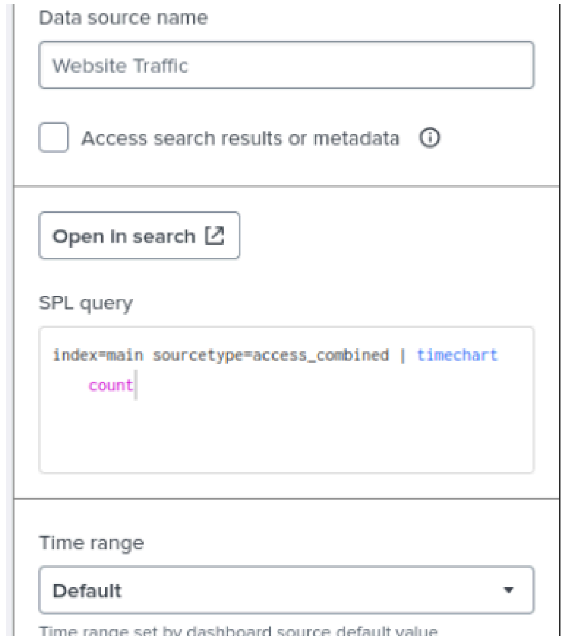
Using **sudo dpkg -i splunk*.deb** to install the package onto the vm. Using **sudo /opt/splunk/bin/splunk start --accept-license --run-as-root** to accept the license agreement and start the splunk service



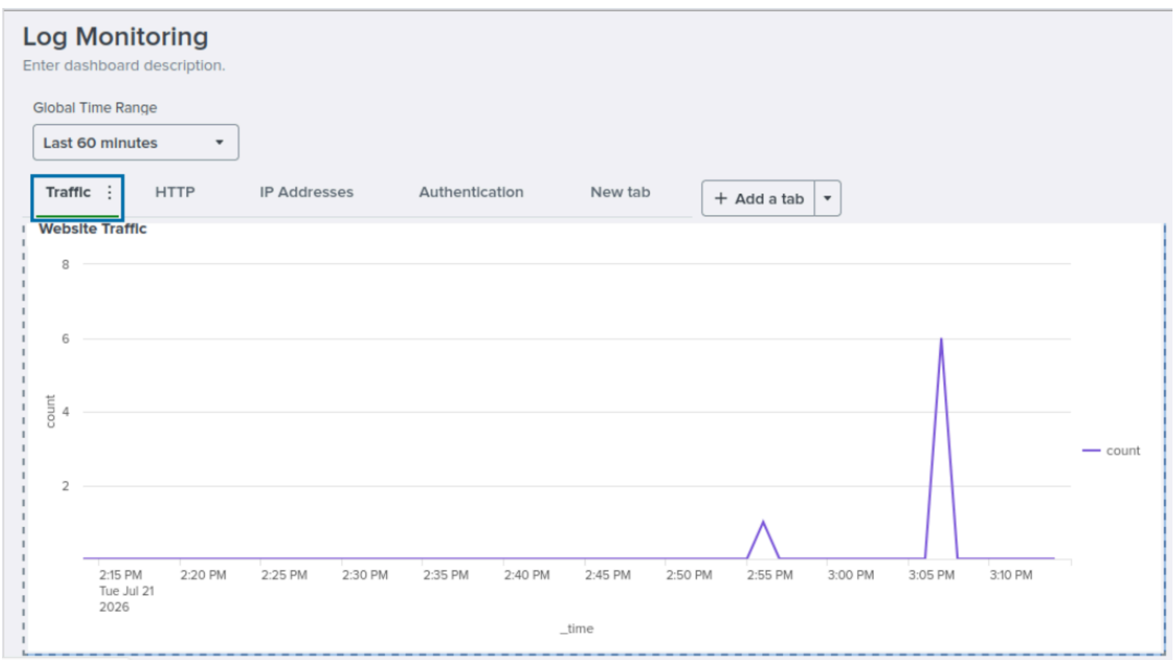
Sign in page at http://shea134:8000

Creating a Dashboard and Panel Inside Splunk

The dashboard was created by going into the **Dashboard** tab to the left and then by clicking on **Create a New Dashboard.**



Web Traffic panel configuration for the Log Monitoring dashboard. This process will be repeated until all of the panels are created while modifying the SPL query to match what is needed for each visualization.



Finished dashboard with panels